Introduction to Computer Networks

A Foundational Guide for Beginners

Chapter 1: What Is a Computer Network?

A computer network is a collection of two or more devices connected together so they can share resources and exchange data. These devices, often called nodes, can include computers, smartphones, printers, servers, and specialized hardware such as routers and switches. Networks range enormously in scale, from two laptops connected in a home office to the global internet, which links billions of devices across every continent. The fundamental purpose of any network is communication: allowing one device to send information to another reliably, even when they are built by different manufacturers and run different software.

Networks are typically classified by their geographic scope. A Local Area Network, or LAN, covers a small physical area such as a single building or campus, and is commonly used in homes and offices. A Wide Area Network, or WAN, spans much larger distances, often connecting multiple cities or countries; the internet itself is the largest example of a WAN. A Metropolitan Area Network, or MAN, sits between these two in scale, typically covering a city. Understanding this classification matters because the technology, cost, and speed of a network often depend heavily on how large an area it needs to cover.

Chapter 2: The OSI Model

The Open Systems Interconnection model, known as the OSI model, is a conceptual framework used to describe how data travels through a network. It divides network communication into seven distinct layers, each responsible for a specific part of the process. Starting from the bottom, these layers are: Physical, Data Link, Network, Transport, Session, Presentation, and Application. Each layer only communicates directly with the layers immediately above and below it, which allows engineers to design and troubleshoot one layer without needing to fully understand every other layer.

The Physical layer deals with the actual transmission of raw bits over a physical medium, such as electrical signals through copper cable or light pulses through fiber optics. The Data Link layer organizes these raw bits into structured frames and handles error detection between directly connected devices. The Network layer, where the Internet Protocol operates, is responsible for addressing and routing data across multiple networks so it can reach a destination that may be many hops away. The Transport layer, home to protocols like TCP and UDP, ensures data arrives completely and in the correct order, or in some cases prioritizes speed over reliability. The remaining three layers, Session, Presentation, and Application, handle higher-level concerns such as maintaining a conversation between two programs, formatting and encrypting data, and providing the actual interface that end-user applications interact with.

Chapter 3: IP Addresses and Subnetting

Every device on a network needs a unique identifier so that data can be correctly delivered to it, much like a postal address identifies a specific house. This identifier is called an IP address. The most common version in use historically, IPv4, represents an address as four numbers between 0 and 255, separated by dots, such as 192.168.1.1. Because IPv4 only supports about 4.3 billion unique addresses, and the number of internet-connected devices has vastly exceeded that, a newer standard called IPv6 was introduced. IPv6 addresses are much longer and written in hexadecimal, providing an effectively unlimited supply of unique addresses for the foreseeable future.

Subnetting is the practice of dividing a large network into smaller, more manageable sub-networks. This is typically done for organizational and security reasons: for example, a company might place its finance department on one subnet and its engineering department on another, so that traffic and access can be controlled separately between them. A subnet mask, working alongside the IP address, determines which portion of the address identifies the network itself and which portion identifies the specific device within that network. Proper subnetting is one of the most practical skills a network administrator needs, since a poorly planned subnet can lead to wasted addresses or an inability to add new devices later.

Chapter 4: DNS - The Domain Name System

Humans are much better at remembering names than long strings of numbers, which is why the Domain Name System, or DNS, exists. DNS acts as a kind of phone book for the internet, translating human-readable domain names such as example.com into the numerical IP address that computers actually use to locate each other, such as 93.184.216.34. Without DNS, users would need to memorize the exact IP address of every website they wanted to visit, which would be impractical at the scale of the modern internet.

When a user types a domain name into their browser, their device sends a DNS query to a DNS resolver, which may check its own cache first before asking a series of authoritative DNS servers to find the correct IP address. This entire process, called DNS resolution, usually completes in a fraction of a second, though the result is often cached locally afterward so that repeated visits to the same site do not require the lookup to be repeated. DNS failures are among the most common causes of a website appearing to be down even when the underlying server is actually running fine, since users cannot reach a server whose address they cannot resolve.

Chapter 5: How the Web Works - HTTP and HTTPS

The Hypertext Transfer Protocol, or HTTP, is the foundational protocol used to transfer web pages and other resources between a browser and a web server. When a browser requests a page, it sends an HTTP request specifying what it wants, and the server replies with an HTTP response containing the requested content along with a status code indicating whether the request succeeded. A status code of 200 means success, 404 means the requested resource was not found, and 500 indicates an error on the server itself. These status codes are an essential tool for developers diagnosing why a website is not behaving as expected.

HTTPS is the secure version of HTTP, and today the vast majority of websites use it. The added 'S' stands for Secure, and it works by wrapping regular HTTP traffic inside an encrypted connection using a protocol called TLS, or Transport Layer Security. This encryption prevents anyone intercepting the traffic, such as someone on the same public Wi-Fi network, from reading the actual content being exchanged, including sensitive information like passwords or credit card numbers. Browsers typically display a padlock icon in the address bar to indicate that a connection is secured with HTTPS, and most modern browsers now actively warn users when they visit a site that only offers plain, unencrypted HTTP.

Chapter 6: Network Security Fundamentals

As networks have grown more central to daily life, protecting them from unauthorized access and malicious activity has become critical. A firewall is one of the most basic and widely used security tools; it monitors incoming and outgoing network traffic and blocks or allows it based on a defined set of rules. Firewalls can be implemented as physical hardware devices sitting at the edge of a network, or as software running directly on an individual computer.

A Virtual Private Network, or VPN, creates an encrypted tunnel between a user's device and a remote server, making it appear as though the user's traffic is originating from that remote server's location. This is commonly used by remote employees to securely access their company's internal network, and by individuals who want to protect their traffic from being observed on untrusted networks, such as public Wi-Fi in a coffee shop. Common threats that network security aims to defend against include malware, which is malicious software designed to damage or gain unauthorized access to a system, and phishing, a social engineering technique where attackers trick users into revealing sensitive information by impersonating a trustworthy source, often through email.